Processor/Bricklet

U101A
XMC1404
VDDP
VSSP
VDD
VSS
100nF
220nF
Boot
BOOT
J101
blue
D103
"status led"
1k
R124
U101B
P0.0_IO
P0.1_IO
P0.2_IO
P0.3_IO
P0.4_IO
P0.5_IO
P0.6_IO
P0.7_IO
P0.8_IO/RTC_XTAL1
P0.9_IO/RTC_XTAL2
P0.10_IO/XTAL1
P0.11_IO/XTAL2
P0.12_IO
P0.13_IO
P0.14_IO
P0.15_IO
C12816MHz
10pF
X101
SW101
SW102
OS103011MS8QP1
P0_0
P0_5
Maximum Power Setting
no jumper: invalid
P0_0 = 3V3, P0_5 = 3V3 -> invalid
P0_0 = DNC, P0_5 = 3V3 -> 6A
P0_0 = GND, P0_5 = 3V3 -> 10A
P0_0 = 3V3, P0_5 = DNC -> 13A
P0_0 = DNC, P0_5 = DNC -> 16A
P0_0 = GND, P0_5 = DNC -> 20A
P0_0 = 3V3, P0_5 = GND -> 25A
P0_0 = DNC, P0_5 = GND -> 32A
P0_0 = GND, P0_5 = GND -> reserved
Ideal Diode 5V Power for ESP
Q105
SSM3J355R,LF
U103
DMMT5401
10k R140
47k R153
FB
L103
P101
CON-SENSOR2
RP101
82C101
220pF
S-CS
S-CLK
S-MOSI
S-MISO
U101C
P1.0_IHO
P1.1_IHO
P1.2_IHO
P1.3_IHO
P1.4_IHO
P1.5_IHO
P1.6_IHO
CP-PWM
INPUT-MOTOR-SWITCH
RELAY
GP-OUT
CP-DISCONNECT
MOTOR_ENABLE
MOTOR_PHASE
U101D
P2.0_AIO
P2.1_AIO
P2.2_AI
P2.3_AI
P2.4_AI
P2.5_AI
P2.6_AI
P2.7_AI
P2.8_AI
P2.9_AI
P2.10/AIO
P2.11/AIO
P2.12/AIO
P2.13/AIO
GP-IN
485-DE
485-REB
485-D
485-R
VPP
VCP1
VCP2
V12
V-12
AC1
AC2
SPI Slave/CH0
P0.13 : USIC0_CH0-DX2F : SEL
P0.14 : USIC0_CH0-DX1A : CLK
P0.15 : USIC0_CH0-DX0B : MOSI
P2.0 : USIC0_CH0-DOUT0 : MISO
U101E
P3.0_IO
P4.4_IO
P4.5_IO
P4.6_IO
P4.7_IO
DC-X6
DC-X30
LED
P112
DC-Protect
DC-ERR
DC-TST
DC-X6
DC-X30
RP103
R1
C107
1uF
5V->3.3V
U110
TPS70933DBVR
C103
L101
C106
C110
2.2uF
C111
100uF
Ersatz TPS70933QDBVRQ1

AC/DC

P104
Power
P106
L'+PE
F102
FUSE
D115
1N4007
D116
1N4007
R118
RNF12FTD249K
R116
RNF12FTD249K
U109
HCPL2731
RP105
10k
P103
F101
FUSE
R128
MOV-10D391KTR
Phoenix
1755736
1757019
U111
IRM02-12
C105
4.7uF/50V
C124
C125
U112
WPMDL110
VIN
VOUT
GND
C126

RS485

RP102
10k
U113
SP3491
R
REB
DE
D
VCC
GND
A
B
Z
Y
NC
C132
100nF
750
R134
DNP
R135
750
R136
SMAJ24CA
SMAJ24CA
D111
L104A
L104B
L104C
L104D
WE-SL
R137
120
P110
MODBUS
PTC 0.45A
PTC101
0ZC.0016FF2E

CP/PP

L102
NR3015T6R8M
U102
PEM1-S12-D12-S
VIN
GND
-Vo
0V
+Vo
C104
4.7uF/50V
C108
4.7uF/50V
C109
4.7uF/50V
-12V
+12V
Vcpadc = (75*Vcp+990)/600
Vcpadc = Vcp-300k/100k*Vcpadc+300k/75k*3.3V-300k/75k*Vcpadc
Option
0402 1NF 100V COG
R125
75k
R126
300k
R127
100k
C120
DNP
R110
75k
R108
300k
R109
100k
C121
DNP
R105
100k
R106
100k
R103
10k
100nF/30V
C112
OPA171AIDBV
R107
910
öffner
CPOUT
CPOUT2
SMF4L13CA
D105
100nF/30V
C113
P105
CP/PP/PE
PE
CP
PP
R111
1k
R112
10k
R141
2k
D107
PESD3V3L1BA
Rpp = (VPP*1k*2k)/(5V*2k-VPP*(1k+2k))
R100
300k
R129
75k
R102
100k
R130
300k
R104
100k
R122
REL101
AGQ200S12Z
D112
B0520LW-7
Q103
2N7002P
R138
1k

12V Input/Motor/GP

C116
100nF/30V
C117
100nF/30V
U106
CP1
CP2
VCP
FAULTn
RESETn
PHASE
ENABLE
MODE
OCLSEL
VREF
VBB
OUTA
OUTB
AIOUT
LSS
GND
PAD
EP
A5950GLP
C118
100nF/30V
C119
100nF/30V
C114
10uF
12V
RP104
1k
MOTOR_FAULT
MOTOR_ENABLE
MOTOR_PHASE
SMF4L13CA
D108
SMF4L13CA
D109
P107
Motor
R119
10k
R117
1k
D104
P109
Enable
R131
1k
R132
10k
D102
P113
GP
R123
1k
R139
10k
Q104
2N7002P
D113
D114
P108
Switch_LED
Q101
2N7002P
D101
R120
1k
R121
10k
D110

Relay

P102
135° 1884340000
180° 1825960000
Relay
R113
MOV-10D391KTR
AQH3213AX
U108
R114
120
Q102
SI2305CDS-T1-GE3
R115
1k

Rail DRILL DRILL
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Sheet: /
File: working.kicad_sch

Title: EVSE Bricklet

Size: A3	Date: 2023-01-20	Rev: 2.0.6
KiCad E.D.A.	kicad-cli 7.0.6-7.0.6-ubuntu22.04.1	Id: 1/1

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